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A Crowdsensing-based Approach for Footbridge Frequency Identification using Pedestrian Response

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Abstract

Indirect bridge health monitoring methods using vehicle response have attracted growing attention from researchers worldwide. However, due to the load and access restrictions, such vehicle-based methods are generally not applicable to footbridges, which represent a significant portion of modern transportation infrastructure. The primary users of footbridges are pedestrians whose dynamic interactions with the structure provide a potential source of measurable vibration data. This paper proposes a novel crowdsensing-based approach for footbridge frequency identification using response collected from pedestrians performing several activities on the footbridge. It provides a paradigm of using collected signals from humans for structural health monitoring (SHM) purposes via crowdsensing strategies. The paper first introduces the fundamental pedestrian-footbridge interaction models and then details the key processes of the proposed approach, including data collection, data processing, and frequency identification. Numerical simulations are conducted to validate the feasibility and robustness of the method. The results demonstrate that pedestrian-borne data can effectively capture the dynamic properties of footbridges, paving the way for a low-cost, scalable, and human-centric SHM strategy.

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1. Introduction

In recent decades, structural health monitoring (SHM) has attracted increasing attention from researchers and engineers due to the deterioration and aging of existing structures (Limongelli, 2020; Zhang et al., 2023). Unlike traditional approaches that rely heavily on visual inspections conducted by experienced engineers, SHM employs sensor-acquired data and advanced data processing techniques to support decision-making for stakeholders. Through SHM techniques, reliable information can be obtained for anomaly detection, damage diagnosis, early warning, and appropriate maintenance planning (Worden and Manson, 2007). Among various SHM approaches, vibration-based methods have shown promising performance because of their robustness and convenience in engineering applications (Hou and Xia, 2021; Li et al., 2025).

Bridges, as a critical component of transportation systems, play an essential role in daily human activities. In recent years, researchers have investigated the use of sensors attached to bridge surfaces or embedded within structural components to measure bridge responses for condition assessment. However, several limitations have been identified when sensors are directly installed on bridges (referred to as the direct method). For instance, many sensors are often required to capture bridge characteristics, and maintaining such sensor networks over long periods can be costly. As a result of these high costs, the direct method has mainly been applied to important bridges, such as long-span and large-scale bridges. In contrast, short-span bridges and footbridges, despite being essential parts of transportation networks worldwide, have received relatively limited research attention over the past decades.

To address these issues, the indirect method was proposed by Yang et al. (2004). In this approach, sensors are installed on vehicles passing over bridges. Through the interaction between the vehicle and the bridge, bridge-related information can be effectively transferred to the vehicle. Data processing techniques are then applied to extract key bridge information from the vehicle response, such as modal parameters including natural frequencies, mode shapes, and damping ratios. Over the past decades, the indirect method has been significantly developed (Li et al., 2026b). Because sensors are installed on vehicles rather than directly on bridges, the identification of bridge information is strongly affected by vehicle properties and road roughness (Xu et al., 2024). To reduce the influence of these external factors, such as vehicle characteristics, road roughness, and environmental noise, crowdsensing approaches have been introduced for indirect bridge health monitoring (Lan et al., 2024). These methods typically employ multiple vehicles with different properties and rely on existing sensing tools, such as smartphones or onboard sensors, without requiring additional sensors (Matarazzo et al., 2018; Stoura et al., 2025), which can take advantage of crowdsensing data. Typically, crowdsensing data is collected by different devices movable around infrastructures. Then, the collected data will be automatically uploaded to cloud platform. In the cloud, data analysis will then be performed to extract key features of the data, which help with assessing the health condition of infrastructures and supporting decision-making for engineers (Mei et al., 2020). Nevertheless, current indirect monitoring methods using crowdsensing data are mainly applied to bridges that support heavy vehicles. Due to load limitations, they are generally not suitable for footbridges.

In recent years, researchers have extended the use of micromobility for footbridge condition assessment. Micromobility typically includes bicycles, electric scooters, and other permitted human-powered vehicles. These studies exploit the crowdsensing potential of smartphones on human-powered vehicles to identify the modal parameters of lightweight footbridges. For example, Quqa et al. (2022) developed a foundational framework to identify footbridge natural frequencies and operational deflection shapes, demonstrating that the high stiffness and low mass of bicycles can reduce adverse vehicle-bridge interaction and engine-related noise. Subsequently, Massarelli et al. (2026) expanded this framework by introducing an automated operational modal analysis approach based on the data-driven Stochastic Subspace Identification (SSI-DATA) algorithm combined with DBSCAN clustering. This method enabled automated estimation of footbridge natural frequencies and damping ratios. By applying singular value decomposition to isolate modal components, the study also achieved automated extraction of operational deflection shapes. Experimental validation on a steel tied-arch footbridge in Bologna, Italy, confirmed the robustness of the proposed approach. However, field tests conducted by May et al. (2024) showed that rider-bicycle dynamics, such as pedaling cadence and posture, can introduce nonstationary frequency shifts and resonance. To mitigate these effects, Li et al. (2024) employed electric scooters and proposed a smartphone-based footbridge damage detection method using scooter vibration data and convolutional neural networks (CNNs). Time-frequency representations processed by a 2D CNN enabled robust damage detection and classification in both simulations and field tests, outperforming 1D

spectral CNNs under noisy and varying operational conditions. Indeed, pedestrians are the primary users of footbridges. Although human-structure interaction (HSI) has been studied for many years (Van Nimmen et al., 2017; Gallegos-Calderón et al., 2021), the use of human response for extracting bridge information has rarely been explored, despite its considerable potential (Li et al., 2026a; Ozer and Feng, 2017). Human response data have been widely used in human activity recognition studies (Wang et al., 2020). Employing pedestrian responses collected by smartphones therefore offers strong potential for crowdsensing-based footbridge health monitoring in the future.

This paper will focus on modeling crowdsensing using response from pedestrians for footbridge frequency identification. The proposed approach eliminates the need for dedicated sensing systems installed on the footbridge and relies solely on data collected by devices such as smartphones carried by pedestrians. Response from pedestrians performing various activities, including standing, jumping, and walking, are analyzed. Numerical simulations are conducted to validate the proposed approach.

The remainder of this paper is organized as follows. Section 2 briefly introduces the theory of pedestrian–footbridge interaction and the models used. Section 3 presents the numerical simulations and the corresponding parameters. Section 4 reports and discusses the results obtained from the numerical simulations. Finally, Section 5 provides the conclusions and outlines future work.

2. Fundamental theories

2.1. Pedestrian-footbridge interaction models in crowdsensing

This section will introduce the basic pedestrian-footbridge interaction models utilized in the crowdsensing-based approach. When pedestrians perform activities such as walking, jogging or simply standing on a footbridge, the bridge vibrates due to both pedestrian-induced loads and ambient excitation. Through the pedestrian's legs, the dynamic information of the footbridge is transferred to the pedestrian body. Then such information can be recorded by wearable devices such as smartphones and/or smartwatches.

The basic biomechanical model for the pedestrian is considered, as shown in Fig. 1. The model consists of a centered mass, a damper and a spring. Suppose that the static mass of the pedestrian is m_{p0} while the parameters of the standing and jumping pedestrians are mass m_{ps} , spring stiffness k_{ps} , and spring damping c_{ps} . For the walking pedestrian, the biomechanical model's parameters are m_{pw} , k_{pw} , and c_{pw} . According to Matsumoto and Griffin (2003), the parameters of the standing human being can be represented as Eq. (1).

$$m_{ps} = 1.03m_{p0}, k_{ps} = 1340m_{p0}, c_{ps} = 51.6m_{p0} \quad (1)$$

For jumping pedestrians, it is assumed that the pedestrian remains standing after the jumping activity. The jumping force is modeled as a sum of Gaussian functions (Sim et al., 2008). In this study, the corresponding equations are adopted from Racic and Pavic (2010), in which experiments involving 44 consecutive jumping cycles were conducted.

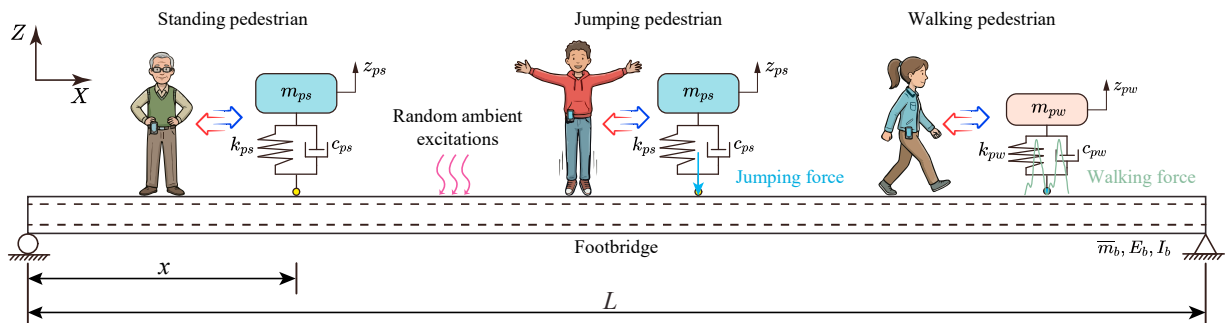


Fig. 1. Pedestrians and models on a footbridge.

According to Da Silva et al. (2013), parameters of the walking pedestrian's biomechanical model can be estimated from Eqs. (2)-(4), which are defined empirically by the authors' based on experimental outcomes. When walking on the structure, the pedestrian applies the walking force to the footbridge. The simulated walking force equations are

adopted from Máca and Štěpánek (2017).

$$m_{pw} = 97.082 + 0.275m_{p0} - 37.518f_{pw}^p \quad (2)$$

$$c_{pw} = 29.041m_{pw}^{0.883} \quad (3)$$

$$k_{pw} = 30351.744 - 50.261c_{pw} + 0.035c_{pw}^2 \quad (4)$$

The footbridge model utilized in this work is a simply supported beam, as shown in Fig. 1, discretized into N finite elements with parameters: mass per unit length $\bar{m}_b$, Young's Modulus E_b , and moment of Inertia I_b . The bridge's span length is L . The equilibrium equation of the footbridge can be written as in Eq. (5),

$$\mathbf{M}\ddot{\mathbf{z}}(t) + \mathbf{C}\dot{\mathbf{z}}(t) + \mathbf{K}\mathbf{z}(t) = \mathbf{p}(t) \quad (5)$$

where $\mathbf{M}$, $\mathbf{K}$, and $\mathbf{C}$, are the mass, stiffness, and damping matrices of the footbridge model. The function $\mathbf{p}(t)$ represents the external force (in this study, it is the pedestrian's interaction force and ambient excitations) applied to the footbridge's nodes. The functions $\ddot{\mathbf{z}}(t)$, $\dot{\mathbf{z}}(t)$, and $\mathbf{z}(t)$ are respectively the vertical acceleration, velocity, and displacement of the footbridge. For the footbridge damping, Rayleigh damping is assumed. The footbridge's modal parameters can be obtained from eigen decomposition.

After establishing the pedestrian and footbridge models, the next step is to couple them. In this study, ambient excitation (f_a), drawn from $N(0,1)$, is considered acting simultaneously to human action. Before the interaction between pedestrians and the footbridge, both the pedestrian and footbridge models are assumed to be static, with zero acceleration, velocity, and displacement. For the standing pedestrian, only random ambient excitation is applied at the beginning. In contrast, for jumping and walking pedestrians, the corresponding jumping and walking forces are applied initially. Under these external excitations, the footbridge begins to vibrate and transfers its vibrations to the pedestrian. The pedestrian then responds accordingly, and the relative displacement and velocity between the pedestrian and the footbridge change accordingly. These changes, in turn, affect the vibration of the footbridge. Through iterative coupling, the difference between the excitations $\mathbf{p}(t)$ in two consecutive iterations gradually decreases and eventually converges to a small threshold, which is set to 10^{-2} N in this study.

2.2. Crowdsensing-based indirect footbridge identification approach

On a footbridge, many pedestrians may be present and perform different activities. Here, responses collected from crowdsensing pedestrians are employed. The proposed crowdsensing-based approach is shown in Fig. 2.

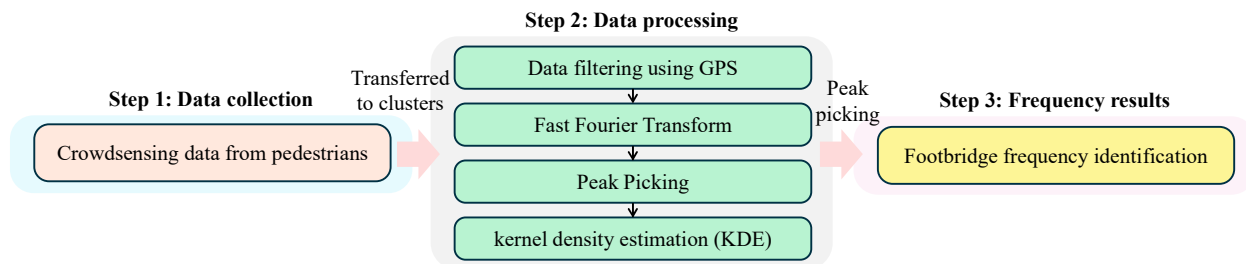


Fig. 2. Proposed crowdsensing-based approach.

In Step 1, crowdsensing data are collected from pedestrians using devices such as smartphones and then transmitted to a cluster. In Step 2, the cluster is used to process the collected data. First, based on GPS information, data corresponding to the period when pedestrians are on the footbridge are selected. Data length has a significant influence on frequency resolution. In this study, only data segments longer than 5 seconds are utilized. If a data segment is relatively long, e.g., longer than 10 seconds, it will be divided into smaller segments, and this procedure is repeated until all data segments fall within a duration of 5-10 seconds. Next, all data segments are transformed into the

frequency domain to obtain their frequency spectra. Given that the data length ranges from 5 to 10 seconds, the resulting frequency resolution is approximately 0.1-0.2 Hz. Subsequently, spectral peaks are identified from each frequency spectrum using the Peak Picking method, and corresponding frequency values of all peaks are collected. Kernel density estimation (KDE) is then applied to these frequency values to obtain a density curve of frequencies. In Step 3, the footbridge frequencies are identified by selecting the dominant peaks of the KDE curve.

3. Numerical simulations

This section introduces the parameters used in numerical simulations of crowdsensing pedestrians on a footbridge. According to Walpole et al. (2012), the region-based average human body mass is 62 kg, with a standard deviation of approximately 6.6 kg. Therefore, in this study, the mass of pedestrians are drawn from a normal distribution $N(62, 6.6)$. The parameters of the biomechanical model of a standing pedestrian are derived from Eq. (1), while that of a walking pedestrian is from Eqs. (2)-(4).

The footbridge parameters are based on Li et al. (2024), considering a span length $L = 33$ m, flexural stiffness $E_b I_b = 1.389 \times 10^9 \text{ N} \cdot \text{m}^2$, and mass per unit length $\bar{m}_b = 224 \text{ kg/m}$. The footbridge is discretized into $N=20$ finite elements. The damping ratios of the first two modes are set to $\xi_1 = \xi_2 = 0.02$ (Li et al., 2023). Based on these parameters, the first three natural frequencies of the footbridge are calculated as $f_{b1} = 3.593 \text{ Hz}$, $f_{b2} = 14.373 \text{ Hz}$, and $f_{b3} = 32.353 \text{ Hz}$. Three pedestrian activities, standing, jumping, and walking, are considered in the crowdsensing network.

On a typical day, pedestrians perform various activities on the footbridge. In general, more pedestrians walk across the footbridge, while fewer pedestrians stand or jump on it. In this study, an approximate ratio of 5:3:2 is adopted for walking, standing, and jumping pedestrians, respectively. The numbers of pedestrians are treated as integers drawn from different uniform distributions, as defined in Eq. (6). Standing and jumping pedestrians are assumed to be located at different positions along the footbridge. For walking pedestrians, four representative speeds, $v_p = 1.1 \text{ m/s}$, 1.5 m/s , 2.2 m/s and 3.3 m/s , corresponding to slow walking, normal walking, fast walking, and jogging, respectively, are considered, as reported in Máca and Štěpánek (2017). Furthermore, 5% white noises are added to the collected pedestrians responses (Li et al., 2023).

$$X_w \sim U_d(25,50), X_s \sim U_d(15,30), X_j \sim U_d(10,20) \quad (6)$$

Pedestrian activities on the footbridge are then simulated over a period of 30 days, representing one month. The numbers of pedestrians performing different activities during this period are shown in Fig. 3.

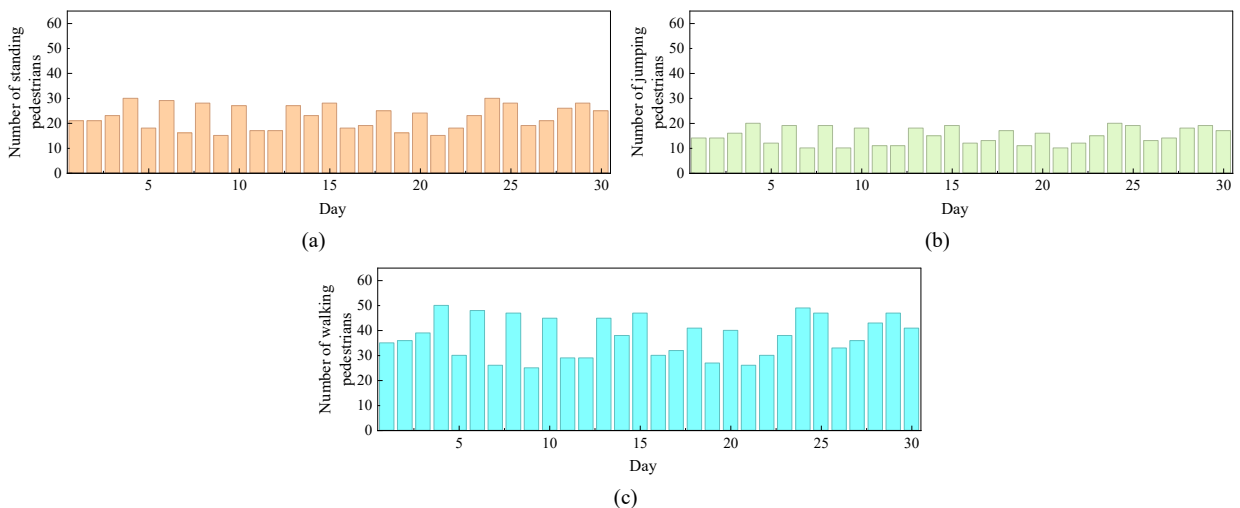


Fig. 3. Number of pedestrians in a month: (a) standing; (b) jumping; and (c) walking.

4. Results and discussions

In the proposed method, pedestrians with diverse parameters are present on the footbridge. When interacting with the bridge, the information related to the bridge, that is reflected in the pedestrians' response, should remain consistent. This principle forms the basis for identifying footbridge frequencies using pedestrian response data.

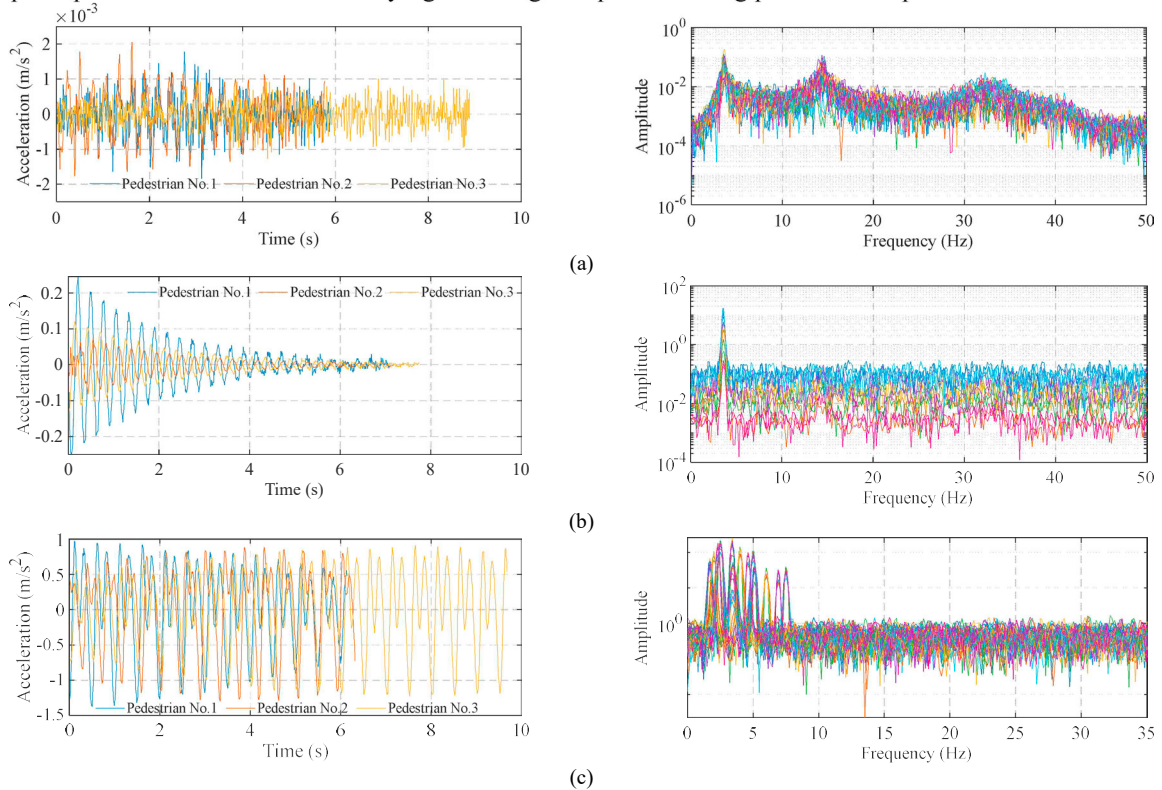


Fig. 4. Response and frequency spectra of pedestrians on day 1: (a) time history response of 3 standing pedestrians and frequency spectra of 21 standing pedestrians; (b) time history response of 3 jumping pedestrians and frequency spectra of 14 jumping pedestrians; (c) time history response of 3 walking pedestrians and frequency spectra of 35 walking pedestrians.

Fig. 4 shows the response and corresponding frequency spectra of pedestrians on day 1. For clarity, only time history responses of three standing, jumping, and walking pedestrians are plotted whereas the spectra refer to the entire day. It is evident that jumping and walking activities significantly increase acceleration amplitudes. From the frequency spectra of standing pedestrians (Fig. 4a), the first two footbridge frequencies can be well observed, while the third footbridge frequency is notably attenuated. In contrast, the frequency spectra of jumping pedestrians make it difficult to distinguish the second and third footbridge frequencies. For the frequency spectra of walking pedestrians, many peaks appear in the low-frequency range (below 10 Hz). Owing to variations in pedestrian mass, walking speed, signal length, and other factors, the excitation frequencies induced by pedestrians vary considerably. Also, higher footbridge frequencies cannot be identified when only walking pedestrian response is used.

In the cluster, we do not classify pedestrian activities using the transferred data. As a result, the specific activity associated with each transferred data segment is unknown. In the next step, all collected data are transformed into the frequency domain, and peaks are extracted from the frequency spectra of all pedestrian responses. This process yields a set of peak frequencies from all pedestrians. Kernel density estimation (KDE) with a bandwidth of 0.1 Hz, corresponding to the finest frequency resolution, is then applied. The results are shown in Fig. 5. It can be observed that, even using pedestrian responses from a single day, the first two natural frequencies of the footbridge can be identified as the three dominant peaks; however, due to the influence of noise, it can be challenging to clearly pick the third frequency of the footbridge. Instead, when more crowdsensing data (in 1 week or 1 month) is utilized, the influence of noise is weakened, and the second and third footbridge frequencies are highlighted.

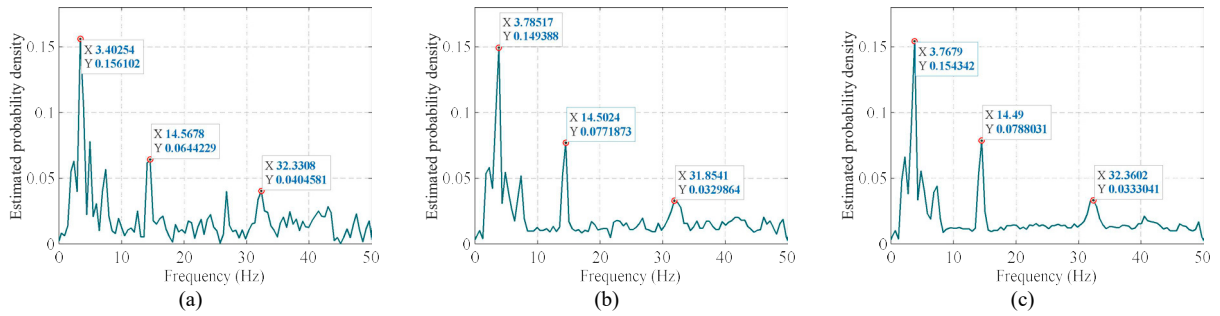


Fig. 5. Footbridge frequency identification results: (a) pedestrians on 1 day; (b) pedestrians in 1 week; (c) pedestrians in 1 month.

To evaluate the accuracy of footbridge frequency identification, the identified frequencies are summarized in Table 1. The results show that when pedestrian data from a single day is used, the first three footbridge frequencies can already be identified with errors within 0.5 Hz. The error is relatively high but by increasing the volume of signals it should decrease. As the volume of crowdsensing data increases, for example, over one month, the identification errors for the footbridge frequencies decrease. The results indicate that increasing the amount of crowdsensing data can lead to more accurate frequency identification.

Table 1. Identification results of footbridge frequency values in different periods.

Modal order	True value	Data on 1 day	Absolute Error	Data in 1 week	Absolute Error	Data in 1 month	Absolute Error
1	3.595 Hz	3.403 Hz	0.19	3.785 Hz	0.19	3.768 Hz	0.17
2	14.373 Hz	14.568 Hz	0.20	14.502 Hz	0.13	14.490 Hz	0.12
3	32.353 Hz	32.331 Hz	0.02	31.854 Hz	0.50	32.360 Hz	0.01

5. Conclusions and future work

This paper proposed a crowdsensing-based framework for identifying footbridge natural frequencies using responses collected from random pedestrians. By exploiting pedestrian-footbridge interaction and aggregating heterogeneous human-borne data, the method enables indirect footbridge frequency identification without installing sensors on the structure. Numerical simulations demonstrated the feasibility and robustness of the approach under diverse pedestrian activities and parameters. The main conclusions can be summarized as follows:

- (1) Pedestrian-borne responses, collected during standing, walking, and jumping activities, can capture key dynamic characteristics of footbridges through pedestrian–structure interaction
- (2) The proposed approach effectively aggregates heterogeneous crowdsensing data and enables reliable identification of multiple footbridge natural frequencies without activity classification.
- (3) Acceptable frequency identification precision can be achieved using a limited amount of data (e.g., one week), while further increasing data volume does not necessarily improve results due to increased randomness.

Future work will focus on incorporating coupling effects between multiple pedestrians and validating the approach through field tests with real smartphone or wearable data. Integration with large-scale participatory sensing and smart city monitoring systems will also be investigated.

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